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VIEWS

1 What is a VIEW in SQL?

A View is a virtual table created from one or more tables using a query

- It shows data from the original table
- It does not store the actual data permanently like a normal table
- whenever we access the view, it shows data from its underlying table or tables

Think of a view as a saved SQL query that we can use like a table

2 Why do we use VIEWS?

- Hide complex queries: you can save a complicated query and access it more easily
- Improve security: you can show users only the data they need
- Simplify data access: you can retrieve data from a view without rewriting the original query

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- Restrict access to specific columns: for ex, show employee names and departments without exposing their salaries

3 How to create a VIEW?
Lets use this employee table as an ~~exp~~ example

emp_id	emp_name	department	Salary
1	Arun	HR	25,000
2	Priya	IT	40,000
3	Rahul	Sales	30,000

Suppose we want a view that display only employee names and departments

Syntax: CREATE VIEW View_name AS
SELECT column1, column2
FROM table_name
WHERE condition;

Ex: CREATE VIEW emp_view AS
SELECT emp_name, department
FROM employees;

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This creates a view named emp-view
The view displays

emp-name	department
Arun	HR
Priya	IT
Rahul	Sales

Salary and emp_id is not included in the view

4 How to access a VIEW?
once we create a view, we can retrieve its data using SELECT

Syntax: SELECT * FROM view_name;
Ex: SELECT * FROM emp_name;

Result:

emp-name	department
Arun	HR
Priya	IT
Rahul	Sales

Here, * means all columns available in the view, not all columns in the original employee table

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5 How to create a VIEW with a WHERE condition?
Suppose we want to display only employees earning more than ₹30000

Ex: CREATE VIEW high_salary_view AS
SELECT emp_name, salary
FROM employee
WHERE Salary > 30000;

Now access the view;
SELECT * FROM high_salary_view;

Result: emp_name Salary
Priya 40000

6 How to DROP a View?
The DROP VIEW command removes the view

Syntax: DROP VIEW view_name;
Ex: DROP VIEW emp_view;

This deletes the view named emp_view, but it does not delete the original employee table or its data

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7 Difference between VIEW and TABLE

VIEW	TABLE
• Virtual table	• Physical table
• Stores the query definition, not the actual result data in a standard view	• Stores actual data
• Generally requires less storage for its definition	• Requires storage for its data
• Data comes from the underlying table	• Data is stored directly

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